CE902 Job Application & CDP Worksheet

Assignment 1: Job Application

LIST OF 5 JOBS FOUND

Job title: Machine Learning Engineer **Company:** DeepMind (Google) **URL Link (if available):** https://www.deepmind.com/careers
Job title: AI Research Scientist **Company:** Microsoft Research Cambridge **URL Link (if available):** https://www.microsoft.com/en-us/research/lab/microsoft-research-cambridge/
Job title: Data Scientist - NLP **Company:** Amazon Web Services (AWS) **URL Link (if available):** https://www.amazon.jobs/en/teams/amazon-web-services
Job title: Computer Vision Engineer **Company:** Meta AI Research **URL Link (if available):** https://ai.meta.com/careers/
Job title: Applied AI Engineer **Company:** OpenAI **URL Link (if available):** https://openai.com/careers

JOB APPLICATION FORM

PERSONAL DETAILS

Name: HRITHIK MAJUMDAR	**Family Name**: SHIBU
Address: D-4-3-D, Haven Road, The Maltings **Postcode**: CO2 8FU **Mobile**: +447920765667 **Email**: hrithikworks22@gmail.com	
Position applied for: Machine Learning Engineer	
Job Ref No.: ML-ENG-2026-001	

EDUCATION

Institution: Shahjalal University of Science and Technology	**Dates**: 19 January, 2019-28 February, 2024
Qualifications: Bachelor of Science (Engineering) in Computer Science and Engineering	

Institution: University of Essex	**Dates**: 03 October, 2025 - present
Qualifications: Master of Science in Artificial Intelligence	
Institution: Coursera (Stanford University)	**Dates:** June 2023 - August 2023
Qualifications: Deep Learning Specialization Certificate	

WORK HISTORY

Employer: R. P. Shaha University	**Dates**: 01 September, 2024 - 26 September, 2025
Job title: Lecturer	
Details: ● Preparing the course outline, lesson plan, and tutorial, and conducting class lectures. ● Conducting and supervising students' projects and research studies. ● Supervising and guiding students' work and activities inside and outside the classroom.	
Employer: Golden Coaching Centre	**Dates**: Dec 2023 – Aug 2024
Job title: Teacher	
Details: ● Conducting regular classes according to the coaching centre schedule. ● Preparing and delivering structured lesson plans, presentations, and study materials. ● Conducting model tests, quizzes, and exam preparations, including solving past papers.	
Employer: Shahjalal University of Science and Technology	**Dates**: April 2019 – Nov 2022
Job title: Data Annotator and Volunteer	
Details: ● Annotated and analysed research projects named 'Emonoba: A dataset for analysing fine-grained emotions on noisy Bangla texts' and 'Bangla Word Stemming'. ● Organised LICT TECH FEST - 2019. ● Volunteered for a conference named ICBSLP - 2019, helped participants to present and publish their papers.	

Why are you interested in working for us?

I am passionate about artificial intelligence research that can be applied towards real-world problems. DeepMind's recent achievement in the field of reinforcement learning with AlphaGo, AlphaFold and AlphaTensor is at the forefront of AI research directly correlating with my career endeavors.

I completed my Master's in Artificial Intelligence from the University of Essex, where I pursued specializations in machine learning & deep learning in domains related to natural language processing and computer vision. My work as an undergraduate researcher on 'Detecting fine-grained emotions on noisy Bangla texts' highlighted my ability to solve pressing problems in low-resource settings, which would be beneficial to DeepMind's diverse research portfolio.

I am particularly drawn to DeepMind because of the company's dedication to building AI responsibly and sharing research openly for the benefit of science as a whole. To work alongside world leading researchers and apply to projects that provide a real societal contribution — from protein structure prediction to energy efficiency — is enormously motivative.

Additionally, teaching at R.P. Shaha University has helped me develop the skill to explain intricate technical ideas explicitly — a key component of many collaborative research settings. I look forward to applying my technical skills, research background and love for innovation in DeepMind's endeavor to solve intelligence in service of science and humanity.

SUPPORTING STATEMENT

Technical Expertise and Research Experience:

I understand machine-learning algorithms deeply, and also that deep learning frameworks (TensorFlow, Keras) and programming languages are core to ML engineering (Python, C++, Java). I completed my masters from Essex featuring intensive courses in neural networks, reinforcement learning and probabilistic modeling. Additionally, having completed Stanford's Deep Learning Specialization earlier in October provided me with a working knowledge of CNNs, RNNs and transformer architectures.

During my undergraduate research, I developed new techniques of emotion detection on noisy Bangla text and constructed a suite of algorithms for word stemming in Bangla text, both datasets showcasing ways to innovate in challenging areas within NLP. These projects involve multiple data cleaning steps, architecture design for model training processes, hyperparameter adjustments and evaluation techniques — all skills used directly in production ML systems.

Teaching and Communication:

Additionally, as a Lecturer at R.P. Shaha University, alongside delivering lectures and courses, I supervised student research projects and guided complex problem solving

activities. This experience has enabled me to hone skills in distilling complex ideas, guiding junior team members and collaborating rapidly — all of which I see as essential for working in multidisciplinary research teams at DeepMind.

Adaptability and Continuous Learning:

As a step to seek further professional growth, I transitioned from being a teacher into embarking on advanced studies in AI. I stay up to date by reading arXiv papers, attending (ICBSLP 2019) conferences, and participating in the academic community through peer review and time as a volunteer. The variety of my maintenance of teaching, research and software development has really nurtured analytical thought, meticulousness as well as aptitude for quickly learning new tech stacks.

I firmly believe that my strong technical foundation, research background, and enthusiasm for AI development make me a great fit for the Machine Learning Engineer role at DeepMind and to contribute significantly to groundbreaking AI pursuits whilst continuing to develop as both a researcher and engineer.

REFERENCES

Dr. Kingkar Prosad Ghosh, Dean and Associate Professor, Dept. of CSE, R. P. Shaha University, Narayanganj, Bangladesh Mobile/cell: +880-1787-122350 Mail: kingkar@rpsu.edu.bd	Mahruba Sharmin Chowdhury, Assistant Professor, Dept. of CSE, Shahjalal University of Science and Technology, Sylhet, Bangladesh Mobile/cell: +880-1917-566699 Mail: mahruba-cse@sust.edu

Assignment 2: Job Application

Background

Continuous Professional Development (CPD) is a structured approach to maintaining and enhancing your skills and knowledge throughout your career. This assignment will guide you through creating a personalised development plan using the provided template. The purpose is to reflect on your current position, set achievable goals, and create a roadmap to achieve professional growth.

"You are a newly hired data scientist at a tech company. Your manager encourages all team members to actively engage in Continuous Professional Development (CPD). Explain how you would plan and record your self-learning activities for the next 12 months to enhance your professional skills. Use examples to illustrate your answer, and reference at least one professional framework or tool to support your plan."

Objective:

1. Enable students to assess their current skills and identify areas for development.
2. Teach the application of CPD principles in planning and tracking professional growth.
3. Foster the habit of lifelong learning by creating actionable and measurable goals.

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NAME:	HRITHIK MAJUMDAR SHIBU	CURRENT POSITION:	MSc Artificial Intelligence Student / Aspiring Machine Learning Engineer
COVERING THE PERIOD FROM:	March 2026	TO: March 2027	
MAIN GOAL:	Transition from academic study to a professional role as a Machine Learning Engineer, developing expertise in production ML systems, advanced deep learning, and collaborative research while establishing a foundation for long-term career growth in AI		

Planned outcome

Where do I want to be by the end of this period? What do I want to be doing?

By the end of 12 Months (by March 2027), I expect to be working as a ML Engineer at a top tech company where I can have an impact on production-grade ML systems and contribute to meaningful experiments. You have at most 9 months to a year (depending on which month you are reading this) to catch up with industry practices before being outcompeted by your peers. So I want to be good at deploying scalable ML in production, adopting MLOps best practices and working with advanced deep learning models like transformers and reinforcement learning.

Within that span of time, I want to have either completed my MSc in Artificial Intelligence with distinction and published at least a research paper or two at one of the esteemed conferences and obtained industry recognized certificates for cloud ML platforms (AWS/GCP/Azure). Finally, I hope to have established a solid professional community through going to conferences, contributing to open-source projects, and participating actively in the AI research society.

Personally, I want to be confidently applying my technical skills to create real cases of problem-solving and mentoring junior developers or students, maintaining a healthy work-life balance that also allows for stepwise learning and knowledge leap. In the end, I want to be seen as an effective and trustworthy ML engineer who parallels research with production environments, makes meaningful contributions to novel AI projects, and grows towards a senior technical track.

What do I want/need to learn?	What will I do to achieve this?	What resources or support will I need?	What will my success criteria be?	Target dates for review and completion
Advanced Deep Learning & Transformer Architectures • Attention mechanisms, BERT, GPT models • Vision Transformers (ViT) • Multimodal learning • Fine-tuning large language models	• Complete online courses • Implement 3-4 transformer models from scratch • Read seminal papers (Attention is All You Need, BERT, GPT series) • Build personal projects using transformer architectures • Participate in Kaggle competitions	• Coursera/Udacity course fees (£200-300) • Cloud GPU credits (AWS/GCP) (£150/month) • Textbooks: 'Transformers for Natural Language Processing' (£40) • HuggingFace API access • University library resources	• Successfully implement and train at least 2 transformer models • Achieve top 10% in 1 Kaggle NLP competition • Complete 3 end-to-end projects with documented code on GitHub • Pass course assessments with >85%	Review: June 2026 Completion: September 2026
Professional Networking & Career Development • Technical interviewing skills • System design for ML systems • Behavioral interview preparation • Industry trends & emerging technologies • Open-source contribution	• Practice 100+ LeetCode problems (Medium/Hard) • Study system design for ML systems • Attend 5+ industry meetups or conferences • Contribute to 2-3 open-source ML projects • Build professional presence (LinkedIn, GitHub, personal website) • Conduct 20+ mock interviews • Join AI/ML communities	• LeetCode Premium (£25/month) • 'Designing Machine Learning Systems' book (£35) • Conference tickets (£100-300 each) • GitHub Pro account (free for students) • Career coaching sessions (£50/session × 4) • Networking event costs (£0-50 per event)	• Successfully pass 3+ technical interviews at FAANG-level companies • Secure job offer as ML Engineer at the target company • 5+ meaningful open-source contributions merged • Professional website with portfolio of 10+ projects • Network of 50+ industry connections • Confident in explaining ML concepts in interviews	Review: Monthly Completion: December 2026
Research Skills & Academic Publishing • Research methodology • Academic writing • Experiment design & reproducibility • Literature review techniques • Conference presentation skills	• Complete MSc dissertation with publishable quality research • Submit 1-2 papers to conferences (ICML, NeurIPS, CVPR, or workshops) • Attend 2 academic conferences (virtual or in-person) • Join research reading groups at the university • Peer review for student conferences • Collaborate with PhD students on research projects	• University supervisor guidance • Conference registration fees (£300-500 per conference) • LaTeX and Overleaf for paper writing • Academic databases: arXiv, Google Scholar, Semantic Scholar • Writing workshops at the university • Research methodology courses	• MSc dissertation completed with distinction (>70%) • At least 1 paper submitted to a peer-reviewed conference • Attended and networked at 2 conferences • Received constructive feedback from peer review • Developed strong literature review skills • Built an academic network in the chosen research area	Review: Every 3 months Completion: June 2027

A brief reflective summary on what you learned from this exercise:

This exercise made me realize that Continuous Professional Development starts not only with ambitious career goals, but defining realistic/structured/measurable actions. Through the layout of my learning needs in technical skills, research and professional growth, I became aware of what I am good at and where those gaps are that will need to be filled before entering industry as a Machine Learning Engineer. It taught me to combine long-term goals, like publishing research and getting a top ML job, with short-term milestones such as taking courses, building projects and refining interview skills.

Professional development must be reviewed, reflected upon, and evidenced; that was another big lesson I learnt. Incorporating success criteria, review dates and resources created a sense of reality about the plan rather than just aspiration. I also learnt that CPD is best when underpinned by frameworks and behaviour-supporting tools for ongoing reflection such as SMART goals or professional development records. This exercise reinforced my belief that career development is a continual cycle of learning, doing, reflecting and adapting — not a one-time endeavor.